

如何编写 51 单片机音乐程序

电路连接很简单, 在 P3.0 端口接个蜂鸣器, 不过效果一般。如果想效果好点, 就加个放大电路, 接个小功率喇叭就行。

音乐程序的设计原理和程序如下:

设计原理

(1) 总体原理:

乐曲中不同的音符, 实质就是不同频率的声音。通过单片机产生不同的频率的脉冲信号, 经过放大电路, 由蜂鸣器放出, 就产生了美妙和谐的乐曲。

(2) 单片机产生不同频率脉冲信号的原理:

1) 要产生音频脉冲, 只要算出某一音频的脉冲 (1/频率), 然后将此周期除以 2, 即为半周期的时间, 利用定时器计时这个半周期的时间, 每当计时到后就将输出脉冲的 I/O 反相, 然后重复计时此半周期的时间再对 I/O 反相, 就可以在 I/O 脚上得到此频率的脉冲。

2) 利用 8051 的内部定时器使其工作在计数器模式 MODE1 下, 改变计数值 TH0 及 TL0 以产生不同频率的方法如下:

例如, 频率为 523Hz, 其周期 $T = 1/523 \text{ S} = 1912 \mu\text{S}$, 因此只要令计数器计时 $956 \mu\text{S} / 1 \mu\text{S} = 956$, 在每计数 956 次时就将 I/O 反接, 就可得到中音 DO (523Hz)。

计数脉冲值与频率的关系公式如下:

$$N = F_i / 2 / F_r$$

(N: 计数值, F_i : 内部计时一次为 $1 \mu\text{S}$, 故其频率为 1MHz, F_r : 要产生的频率)

(3) 其计数值的求法如下:

$$T = 65536 - N = 65536 - F_i / 2 / F_r$$

计算举例:

设 $K = 65536, F = 1000000 = F_i = 1\text{MHz}$, 求低音 DO (261Hz)、中音 DO (523Hz)、高音 DO (1046Hz) 的计数值。

$$T = 65536 - N = 65536 - F_i / 2 / F_r = 65536 - 1000000 / 2 / F_r = 65536 - 500000 / F_r$$

$$\text{低音 DO 的 } T = 65536 - 500000 / 262 = 63627$$

$$\text{中音 DO 的 } T = 65536 - 500000 / 523 = 64580$$

$$\text{高音 DO 的 } T = 65536 - 500000 / 1047 = 65059$$

(4) C 调个音符频率与计数值 T 的对照表如下表所示:

表 9.1 C 调各音符频率与计数值 T 的对照表

音符 频率 (Hz) 简谱码 T 值 音符 频率 (Hz) 简谱码 T 值

低 1 DO 262 63628 #4FA# 740 64860

#1 DO# 277 63731 中 5 SO 784 64898

低 2 RE 294 63835 #5 SO# 831 64923

#2 RE# 311 63928 中 6 LA 880 64968

低 3 MI 330 64103 #6 932 64994

低 4 FA 349 64103 中 7 SI 988 65030

#4 FA# 370 64260 高 1 DO 1046 65058

低 5SO 392 64260 #1DO# 1109 65085
#5SO# 415 64331 高 2RE 1175 65110
低 6LA 440 64400 #2RE# 1245 65124
#6 466 64463 高 3M 1318 65157
低 7SI 494 64524 高 4FA 1397 65178
中 1DO 523 64580 #4FA# 1480 65198

(5) 每个音符使用 1 个字节, 字节的高 4 位代表音符的高低, 低 4 位代表音符的节拍, 下表为节拍码的对照。但如果 1 拍为 0.4 秒, 1/4 拍是 0.1 秒, 只要设定延迟时间就可求得节拍的时间。假设 1/4 节拍为 1DELAY, 则 1 拍应为 4DELAY, 以此类推。所以只要求得 1/4 拍的 DELAY 时间, 其余的节拍就是它的倍数, 如下表为 1/4 和 1/8 节拍的时间设定。

表 9.2 节拍码对照表

1/4 节拍 1/8 节拍
节拍码 节拍数 节拍码 节拍数
1 1/4 拍 1 1/8 拍
2 2/4 拍 2 1/4 拍
3 3/4 拍 3 3/8 拍
4 1 拍 4 1/2 拍
5 1 又 1/4 拍 5 5/8 拍
6 1 又 1/2 拍 6 3/4 拍
7 1 又 3/4 拍 7 7/8 拍
8 2 拍 8 1 拍
9 2 又 1/4 拍 9 1 又 1/8 拍
A 2 又 1/2 拍 A 1 又 1/4 拍
B 2 又 3/4 拍 B 1 又 3/8 拍
C 3 拍 C 1 又 1/2 拍
D 3 又 1/4 拍 D 1 又 5/8 拍
E 3 又 1/2 拍 E 1 又 3/4 拍
F 3 又 3/4 拍 F 1 又 7/8 拍

表 9.3 各调节拍的时间设定表

1/4 节拍 1/8 节拍
曲调值 DELAY 曲调值 DELAY
调 4/4 125 毫秒 调 4/4 62 毫秒
调 3/4 187 毫秒 调 3/4 94 毫秒
调 2/4 250 毫秒 调 2/4 125 毫秒

(6) 建立音乐的步骤:

- 1) 先把吧乐谱的音符找出, 然后由上表建立 T 值表的顺序。
- 2) 把 T 值表建立在 TABLE1, 构成发音符的计数值放在“TABLE”。
- 3) 简谱码(音符)为高位, 节拍为(节拍数)为低 4 位, 音符节拍码放在程序的“TABLE”处。

表 9.4 简谱对应的简谱码、T 值、节拍数

简谱 发音 简谱码 T 值 节拍码 节拍数
5 低 5SO 1 64260 1 1/4 拍

6 低 6LA 2 64400 2 2/4 拍
7 低 7SI 3 64524 3 3/4 拍
1 中 1DO 4 64580 4 1 拍
2 中 2RE 5 64684 5 1 又 1/4 拍
3 中 3M 6 64777 6 1 又 2/4 拍
4 中 4FA 7 64820 7 1 又 3/4 拍
5 中 5SO 8 64898 8 2 拍
6 中 6LA 9 64968 9 2 又 1/4 拍
7 中 7SI A 65030 A 2 又 2/4 拍
1 高 1DO B 65058 B 2 又 3/4 拍
2 高 2RE C 65110 C 3 拍
3 高 3M D 65157 D 3 又 1/4 拍
4 高 4FA E 65178 E 3 又 2/4 拍
5 高 5SO F 65217 F 3 又 3/4 拍
不发音 0

1/4 拍的延迟时间=187 毫秒

```
DELAY:    MOV    R7,#2
D2:       MOV    R4,#187
D3:       MOV    R3,#248
DJNZ     R3,$
DJNZ     R4,D3
DJNZ     R7,D2
RET
```

4. 程序范例

```
ORG 0000H      ; 主程序起始地址
SJMP START    ; 跳至主程序
ORG 000BH      ; TIMER0 中断起 始地址
            LJMP TIM0      ; 跳至 TIMER0 中断子程序
START: MOV TMOD,#01H ; 设 T0 在 M1
            MOV IE,#82H    ; 中断使能
START0: MOV 30H,#00      ; 取简谱码指针
NEXT:  MOV A,30H          ; 简谱码指针载入 A
            MOV DPTR,#TAB ; 至 TAB 取简谱码
MOVC A,@A+DPTR ;
            MOV R2,A      ; 渠道的简谱码暂存于 R2
            JZ  END0      ; 是否渠道 00 (结束码)
            ANL A,#0FH    ; 不是, 则取低 4 位 (节拍码)
            MOV R5,A      ; 将节拍码存入 R5
            MOV A,R2      ; 将取得的简谱码再载入 A
            SWAP A        ; 高低 4 位交换
            ANL A,#0FH    ; 取低 4 位 (音符码)
            JNZ SING      ; 取得的音符码是否为 0?
            CLR TR0       ; 开始, 则不发音
```

```
        SJMP D1    ; 跳至 D1
SING:   DEC A     ; 取到的音符码减 1 (不含 0)
        MOV 22H,A ; 存入 (22H)
        RL  A     ; 乘 2
        MOV DPTR,#TAB1 ; 至 TABLE1 取相对的高位字节计数值
        MOVC A,@A+DPTR ;
MOV TH0,A ; 取到的高位字节存入 TH0
MOV 21H,A ; 取到的高位字节存入 (21H)
MOV A,22H ; 在载入取到的音符码
RL  A     ; 乘 2
INC  A     ; 加 1
MOVC A,@A+DPTR ; 至 TABLE1 取相对的低位字节计数值
MOV TL0,A ; 取到的低位字节存入 TL0
MOV 20H,A ; 取到的低位字节存入 (20H)
SETB TR0  ; 启动 TIMER0
D1:     LCALL DELAY ; 其本单位时间 1/4 拍 187 毫秒
        INC 30H    ; 取简谱码指针加 1
        JMP NEXT ; 取下一个简谱码
END0:   CLR TR0    ; 停止 TIMER0
        JMP START0 ; 重复循环
TIM0:   PUSH ACC   ; 将 A 的值暂存于堆栈
        PUSH PSW   ; 将 PSW 的值暂存于堆栈
        MOV TL0, 20H ; 重设计数值
        MOV  TH0, 21H ;
        CPL  P3.0  ; 将 P3.0 位反相,控制蜂鸣器发声
        POP  PSW   ; 至堆栈取回 PSW 的值
        POP ACC   ; 至堆栈取回 A 的值
        RETI
DELAY:  MOV R7, #02
D2:     MOV  R4, #187
D3:     MOV R3, #248
        DJNZ R3, $
        DJNZ R4, D3
        DJNZ R7, D2
        RET
TAB1:   ; 决定节拍
DW 64260, 64400, 64521, 64580
        DW 64684, 64777, 64820, 64898
        DW 64968, 65030, 65058, 65110
        DW 65157, 65178, 65217
TAB:    ; 乐曲名称《梁祝》
        DB 02H,82H,62H,52H,48H,02H,52H,32H,22H,18H
        DB 83H,91H,72H,62H,51H,61H,71H,61H,83H,61H
        DB 81H,51H,61H,71H,61H,51H,46H,82H,32H,52H
```

DB 22H,42H,16H,21H,41H,18H,0E4H,13H,21H,43H
DB 51H,21H,41H,12H,83H,81H,61H,81H,58H,53H
DB 61H,31H,22H,13H,21H,42H,52H,0E2H,42H,21H
DB 11H,91H,41H,18H,63H,81H,32H,52H,21H,41H,
DB 16H,0E4H,11H,21H,31H,51H,26H,11H,21H,43H
DB 51H,82H,62H,52H,61H,51H,42H,21H,11H,0E4H
DB 44H,21H,41H,21H,11H,0E1H,11H,21H,41H,18H
DB 61H,81H,51H,61H,51H,41H,32H,21H,41H,18H
DB 08H,0H,04H ; 曲子最后静音 5 拍长的时间
DB 00H ; 乐曲结束
END

SONG1: DB 04H ;1=C 2/4 军港之夜

DB
32H,54H,52H,32H,54H,52H,32H,12H,12H,32H,32H,54H,52H,32H,52H,52H,32H,32H,21H,31H,
24H,0E2H,0D1H,0E1H,0D2H,0C2H,14H,14H
DB
0C2H,32H,32H,12H,21H,31H,24H,32H,34H,0D2H,0C2H,14H,14H,32H,52H,52H,32H,52H,54H,
32H,34H,31H,21H,12H,24H,24H
DB
34H,0E2H,0C2H,0D1H,0E1H,0D4H,12H,0E2H,32H,32H,0E2H,0D1H,0E1H,0D4H,0D2H,0E2H,
32H,32H,0E2H,0E2H,0D1H,0E1H,0D4H
DB
22H,0D1H,11H,0E2H,0D2H,0C4H,0C4H,32H,52H,52H,32H,62H,5H,61H,54H,31H,52H,31H,12
H,31H,31H,32H,54H,52H,32H,52H,52H,32H
DB
32H,21H,31H,24H,0E2H,0D1H,0E1H,0D2H,0C2H,34H,34H,32H,52H,52,32H,62H,51H,61H,54
H,31H,52H,31H,12H,32H,12H,32H,32H,54H,52H
DB 32H,52H,52H,31H,31H,32H,21H,31H,24H,0E2H,0D1H,0E1H,0D2H,0C2H,14H,14H
DB 32H,52H,52H,32H,32H,21H,31H,24H,0E2H,0D1H,0E1H,0D2H,0C2H,14H,14H
DB
0C2H,32H,32H,12H,21H,31H,24H,32H,34H,0D2H,0C2H,14H,14H,32H,52H,52H,32H,52H,54H,
32H,34H,31H,21H,12H,24H,24H
DB
34H,0E2H,0C2H,0D1H,0E1H,0D4H,12H,0E2H,32H,32H,0E2H,0D1H,0E1H,0D4H,0D2H,0E2H,
32H,32H,0E2H,0E2H,0D1H,0E1H,0D4H
DB
22H,0D1H,11H,0E2H,0D2H,0C4H,0C4H,32H,52H,52H,32H,62H,5H,61H,54H,31H,52H,31H,12
H,31H,31H,32H,54H,52H,32H,52H,52H,32H
DB

32H,21H,31H,24H,0E2H,0D1H,0E1H,0D2H,0C2H,34H,34H,32H,52H,52,32H,62H,51H,61H,54
H,31H,52H,31H,12H,32H,12H,32H,32H,54H,52H

DB 32H,52H,52H,31H,31H,32H,21H,31H,24H,0E2H,0D1H,0E1H,0D2H,0C2H,14H,14H

DB 32H,52H,52H,32H,32H,21H,31H,24H,74H,0D2H,0C2H,14H,14H,14H,14H

DB 00H

TABLE2:DW 63835,64021,64103,64260,64400,64524,64580,64684,64777

DW 64820,64898,64968,65030,65058,65110,65157,65178

SONG2: DB 02H ;1=C 4/4 死了都要爱

DB 04H,94H

DB 0E8H,0E4H,0E4H,0E8H,0E4H,0F4H,0D4H,0C4H,0B4H,84H,84H,92H,92H,94H,92H,02H

DB 0A4H,0A2H,0B2H,0B4H,0C4H,84H,82H,92H,94H,0A4H

DB 04H,0C2H,0C2H,0D4H,0C2H,0D2H,0D8H,0D4H,0D2H,92H

DB 0E8H,0E4H,0E4H,0E8H,0E4H,0F4H,0D4H,0C4H,0B4H,84H,84H,92H,92H,94H,92H,02H

DB 0A4H,0A2H,0B2H,0B4H,0C4H,0E8H,0D8H

DB 0C8H,0C8H,08H,08H,78H,28H,38H,34H,24H

DB

24H,64H,64H,24H,64H,72H,72H,78H,58H,54H,44H,38H,34H,14H,34H,84H,74H,74H,62H,62H,
0D8H,78H,28H,38H,34H,24H

DB 24H,64H,64H,24H,64H,72H,72H,72H,94H,92H,94H,0A4H,58H,58H,54H,44H ;#音不准

DB 58H,58H,58H,58H,0A8H,0A8H,0A4H,0B4H,0A4H,84H,0A4H,94H,98H,98H,08H

DB

84H,82H,82H,82H,82H,94H,0A2H,94H,92H,82H,74H,72H,74H,72H,72H,72H,92H,91H,91
H,94H,54H,74H,94H

..*****
,,

DB

0A8H,0A4H,0A2H,0B2H,0C4H,0B4H,0A4H,0A4H,0A4H,92H,92H,98H,82H,72H,78H,04H,0A
4H,0A2H,0B2H,0B4H,0C4H,84H,82H,92H,94H,0A4H

DB 04H,0C4H,0D4H,0C4H,0C8H,04H,94H

DB 0E8H,0E4H,0E4H,0E8H,0E4H,0F4H,0D4H,0C4H,0B4H,84H,84H,92H,92H,94H,92H,02

DB

0A4H,0A2H,0B2H,0B4H,0C4H,84H,82H,92H,94H,0A4H,04H,0C2H,0C2H,0D4H,0D8H,0D4H,
0D2H,92H

DB 0E8H,0E4H,0E4H,0E8H,0E4H,0F4H,0D4H,0C4H,0B4H,84H,84H,92H,92H,94H,92H,02

DB

0A4H,0A2H,0B2H,0B4H,0C4H,84H,82H,92H,94H,0A4H,04H,0C2H,0C2H,0D4H,0D8H,0D4H,
0D2H,92H

..*****
,

DB 0A4H,0A2H,0B2H,0B3H,0C4H,0E8H,0D8H

DB 0C8H,0C8H,0C8H,08H

DB 00H

TABLE3:

DW

64898,64968,65030,65086,65135,65158,65199,64260,64400,64524,64580,64684,64777,64862

SONG3: DB 03H ;G=2/4 中华人民共和国国歌

DB

```
04H,04H,0C2H,14H,12H,12H,12H,11H,11H,0C2H,0D1H,0E1H,14H,14H,02H,32H,12H,21H,31
H,52H,51H,51H,54H
DB
32H,31H,31H,12H,11H,31H,52H,51H,31H,24H,24H,24H,64H,54H,24H,34H,52H,34H,52H,32H,
21H,31H,12H,11H,21H,34H,04H
DB
52H,51H,0D1H,12H,12H,32H,31H,31H,52H,51H,51H,22H,22H,22H,0D2H,0D1H,0D1H,24H,22
H,0C2H,14H,12H,12H,34H,32H,32H,54H,54H,54H,54H
DB
12H,11H,31H,52H,51H,51H,64H,54H,32H,31H,11H,52H,52H,52H,32H,02H,12H,02H,0C4H,14
H
DB
0A2H,0A1H,11H,52H,52H,52H,32H,02H,12H,02H,0C4H,14H,0C4H,14H,0C4H,14H,14H,04H
DB 04H,04H,04H,04H
DB 00H
END;
```

```
/*
*****
/*我这有一个程序，你参考一下 */
/* function:唱生日快乐 */
/*
*****
#include<absacc.h>
#include<reg52.h>
#include<intrins.h>
#define uint unsigned int
#define uchar unsigned char
#define led RD
#define key_rec T1
sbit BUZZER=P3^7;//发声音口
uchar code
T0_H[16]={0xFF,0xFB,0xFB,0xFC,0xFC,0xFC,0xFD,0xFD,0xFD,0xFD,0xFE,0xFE,0xFE,0xFE,
0xFE,0xFE};
uchar code
T0_L[16]={0xF1,0x4,0x90,0xC,0x44,0xAC,0x9,0x34,0x82,0xC8,0x06,0x22,0x56,0x85,0x9A,0x
C1};
uchar code
music[32]={8,0,8,9,8,0xB,0xA,0,8,0,8,9,8,0xC,0xB,0,8,0,8,0xF,0xD,0xB,0xA,9,0xE,0,0xE,0xD,0
xB,0xC,0xB,0x0};
uchar code music_L[32]={2,1,1,4,4,4,4,2,1,1,4,4,4,4,2,1,1,4,4,4,4,2,1,1,4,4,4,4,4};
uchar a,b;
uint second,minute;
```

```
long t0count;
void delay_music(uint v)
{
while(v!=0) v--;
}
void delay_music1(uint j)
{
while(j!=0){j--;delay_music(200);}
}
void delay_music2(uint j)
{
while(j!=0){j--;delay_music1(80);}
}
main()
{
//uint i;
uchar k,n,m;
delay_music(100);
TMOD=0x11;
TH0=0xfc;
TL0=0x66;
t0count=0;
EA=1;
ET0=0;
TR0=0;
ET1=0;
TR1=0;
while(1)
{
ET1=1;
TR1=1;
for(k=0;k<32;k++)
{
n=music[k];
m=music_L[k];
a=T0_H[n];
b=T0_L[n];
delay_music2(m);
}
}
}
time0() interrupt 1 using 1
{
TH0=0xFC;
TL0=0x66;
```

```
t0count=t0count+1;
if (t0count==1000)
{
second++;
if (second==60)
{
minute++;
if (minute==60) minute=0;
second=0;}
t0count=0;
}
}
time1 () interrupt 3 using 3
{
TH1=a;
TL1=b;
BUZZER=!BUZZER;
}
```

早说嘛!!

改一下不就行了嘛

下面是汇编写的

```
ORG      0000H
JMP      MAIN
ORG      000BH
JMP      INTT0
ORG      0100H
MAIN:    MOV      SP,#60H
MOV      TMOD,#01H
SETB     ET0
SETB     EA
SETB     TR0
START0:  SETB     INT0
MOV      30H,#00H
NEXT:    MOV      A,30H
MOV      DPTR,#TABLE
MOVC     A,@A+DPTR
MOV      R2,A
JZ       ENDD
ANL      A,#0FH
MOV      R5,A
MOV      A,R2
```

```
        SWAP    A
        ANL     A,#0FH
        JNZ     SING
        CLR     TR0
        JMP     D1
SING:   DEC     A
        MOV     22H,A
        RL      A
        MOV     DPTR,#TABLE1
        MOVC    A,@A+DPTR
        MOV     TH0,A
        MOV     21H,A
        MOV     A,22H
        RL      A
        INC     A
        MOVC    A,@A+DPTR
        MOV     TL0,A
        MOV     20H,A
        SETB    TR0
D1:     CALL    DELAY
        INC     30H
        JMP     NEXT
ENDD:   CLR     TR0
        JMP     START0

INTT0:
        PUSH    PSW
        PUSH    ACC
        MOV     TL0,20H
        MOV     TH0,21H
        CPL     INT0
        POP     ACC
        POP     PSW
        RETI
DELAY:  MOV     R7,#02
DELAY0: MOV     R4,#187
DELAY1: MOV     R3,#248
        DJNZ    R3,$
        DJNZ    R4,DELAY1
        DJNZ    R7,DELAY0
        DJNZ    R5,DELAY
        RET
TABLE:  DB 82H,01H,81H,94H,84H,0B4H,0A4H,04H
        DB 82H,01H,81H,94H,84H,0C4H,0B4H,04H
```

DB 82H,01H,81H,0F4H,0D4H,0B4H,0A4H,94H
DB 0E2H,01H,0E1H,0D4H,0B4H,0C4H,0B4H,04H
DB 82H,01H,81H,94H,84H,0B4H,0A4H,04H
DB 82H,01H,81H,94H,84H,0C4H,0B4H,04H
DB 82H,01H,81H,0F4H,0D4H,0B4H,0A4H,94H
DB 0E2H,01H,0E1H,0D4H,0B4H,0C4H,0B4H,04H
TABLE1: DW 64260,64400,64524,64580,64684,64777,64820,64898
DW 64968,65030,65058,65110,65157,65178,65217
END